

# Environmental Data Analysis: Homework 5

a.a. 2025/2026

Normal distributions are often applied to count data even though count data can only take on positive integer values.

The goal of these exercises is to discuss how good the normal approximation is.

## Exercise 1

- a. Would the normal be a fair approximation to the Poisson curves for small numbers?
- b. How about for the Binomial for small numbers ?

## Exercise 2

- a. Is it possible to choose the parameters so that the Poisson and Binomial to both have the same mean and variance?
- b. If so what is this parameterization?

## Exercise 3

The annual number of earthquakes registering at least 2.5 on the Richter Scale and having an epicenter within 40 miles of downtown Memphis follows a Poisson distribution with mean 6.5.

- a. What is the probability that at least 9 such earthquakes will strike next year?
- b. Use the Gaussian approximation to Poisson to compute an approximate probability and give your assessment of the quality of the approximation.